



Alzheimer Disease

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Alzheimer disease causes progressive cognitive deterioration and is characterized by beta-amyloid deposits and neurofibrillary tangles in the cerebral cortex and subcortical gray matter. Diagnosis is clinical; laboratory and imaging tests are usually done to look for specific findings that suggest Alzheimer disease and to identify other treatable causes of dementia. Treatment is with a combination of supportive measures and pharmacotherapy. Medications include cholinesterase inhibitors, memantine, and anti-amyloid monoclonal antibody therapies in selected patients.

(See also [Overview of Delirium and Dementia](#) and [Dementia](#).)

Alzheimer disease, a neurocognitive disorder, is the most common cause of [dementia](#); it accounts for 60 to 80% of dementias in older adults. In the United States, an estimated 11% of people ≥ 65 years have Alzheimer disease. The percentage of people with Alzheimer disease increases with age ([1](#)):

- Age 65 to 74: 5%
- Age 75 to 84: 13%
- Age ≥ 85 : 33%

The disease is twice as common among females as among males, partly because females have a longer life expectancy. Prevalence in industrialized countries is expected to increase as the proportion of older adults increases.

General reference

1. [2024 Alzheimer's disease facts and figures](#). *Alzheimers Dement* 2024;20(5):3708-3821. doi:10.1002/alz.13809

Etiology of Alzheimer Disease

Most cases of Alzheimer disease are sporadic, with late onset (≥ 65 years) and unclear etiology. Risk of developing the disease is best predicted by age. However, about 5 to 15% of cases are familial; half of

these cases have an early (presenile) onset (< 65 years) and are typically related to specific genetic mutations.

At least 5 distinct genetic loci, located on chromosomes 1, 12, 14, 19, and 21, influence initiation and progression of Alzheimer disease.

Mutations in genes for the amyloid precursor proteins, presenilin I and presenilin II, may lead to autosomal dominant forms of Alzheimer disease, typically with early onset. In affected patients, the processing of amyloid precursor protein is altered, leading to deposition and fibrillar aggregation of beta-amyloid; beta-amyloid is the main component of neuritic (senile) plaques, which consist of degenerated axonal or dendritic processes, astrocytes, and glial cells around an amyloid core. Beta-amyloid may also alter kinase and phosphatase activities in ways that eventually lead to hyperphosphorylation of tau (a protein that stabilizes microtubules) and formation of neurofibrillary tangles.

Other genetic determinants include the apolipoprotein (apo) E (epsilon) alleles. Apo E proteins influence beta-amyloid deposition, cytoskeletal integrity, and efficiency of neuronal repair. Risk of Alzheimer disease is substantially increased (up to 15-fold) in people with 2 epsilon-4 alleles (ApoE4) and may be decreased in those who have the epsilon-2 allele (1).

Vascular risk factors, such as hypertension, diabetes, dyslipidemia, and smoking, can increase the risk of Alzheimer disease. Growing evidence suggests that aggressive treatment of these risk factors as early as midlife can attenuate the risk of developing cognitive impairment in older age.

The relationship of other factors, such as low estrogen levels and heavy metal exposure, to Alzheimer disease has not been established.

Etiology reference

1. [Raulin AC, Doss SV, Trottier ZA, Ikezu TC, Bu G, Liu CC](#). ApoE in Alzheimer's disease: pathophysiology and therapeutic strategies. *Mol Neurodegener* 2022;17(1):72. doi:10.1186/s13024-022-00574-4

Pathophysiology of Alzheimer Disease

The 2 pathologic hallmarks of Alzheimer disease are

- Extracellular beta-amyloid deposits (in neuritic plaques)
- Intracellular neurofibrillary tangles (paired helical filaments)

The beta-amyloid deposition and neurofibrillary tangles lead to loss of synapses and neurons, which results in gross atrophy of the affected areas of the brain, typically starting at the mesial temporal lobe.

The mechanism by which beta-amyloid peptide and neurofibrillary tangles cause such damage is incompletely understood. There are several theories.

The **amyloid hypothesis** posits that progressive accumulation of beta-amyloid in the brain triggers a complex cascade of events ending in neuronal cell death, loss of neuronal synapses, and progressive neurotransmitter deficits; all of these effects contribute to the clinical symptoms of dementia.

A **sustained immune response and inflammation** have been observed in the brain of patients with Alzheimer disease. Some experts have proposed that inflammation is the third core pathologic feature of Alzheimer disease (1).

Glucose metabolism derangement has been shown to play a potentially important role in the development of Alzheimer disease (2).

Prion mechanisms have been identified in Alzheimer disease. In [prion diseases](#), a normal cell-surface brain protein called prion protein becomes misfolded into a pathogenic form termed a prion. The prion then causes other prion proteins to misfold similarly, resulting in a marked increase in the abnormal proteins, which leads to brain damage. In Alzheimer disease, it is thought that the beta-amyloid in cerebral amyloid deposits and tau in neurofibrillary tangles have prion-like, self-replicating properties.

Pathophysiology references

1. [Kinney JW, Bemiller SM, Murtishaw AS, et al](#): Inflammation as a central mechanism in Alzheimer's disease. *Alzheimers Dement (NY)* 4:575–590, 2018. doi: 10.1016/j.trci.2018.06.014
2. [González A, Calfío C, Churrua M, Maccioni RB](#): Glucose metabolism and AD: evidence for a potential diabetes type 3. *Alzheimers Res Ther* 14(1):56, 2022. doi: 10.1186/s13195-022-00996-8

Symptoms and Signs of Alzheimer Disease

Patients with Alzheimer disease have [symptoms and signs of dementia](#).

The **most common first manifestation** of Alzheimer disease is

- Loss of short-term memory (eg, asking repetitive questions, frequently misplacing objects or forgetting appointments)

Other cognitive deficits tend to involve multiple functions, including the following:

- Impaired reasoning, difficulty handling complex tasks, and poor judgment (eg, being unable to manage bank accounts, making poor financial decisions)
- Language dysfunction (eg, difficulty thinking of common words, errors speaking and/or writing)
- Visuospatial dysfunction (eg, inability to recognize faces or common objects)

Symptoms of Alzheimer disease progress gradually but may plateau for periods of time.

[Behavior disorders](#) (eg, wandering, agitation, yelling, persecutory ideation) are common.

Diagnosis of Alzheimer Disease

- Similar to that of other dementias
- Formal mental status examination
- History and physical examination
- Laboratory testing
- Neuroimaging

Generally, the initial approach to the diagnosis of Alzheimer disease is similar to the [diagnosis of other dementias](#). However, despite clinical and specific laboratory and imaging characteristics, a definitive diagnosis of Alzheimer disease can only be confirmed by postmortem histologic evaluation of brain tissue.

Evaluation includes a thorough history, neurologic examination, and cognitive testing (eg, Mini Mental Status Examination) ([1](#), [2](#), [3](#)). Additional studies to support the diagnosis and exclude other etiologies include CSF analysis, laboratory and biomarker testing, and PET imaging.

Clinical criteria can help differentiate Alzheimer disease from other forms of dementia, such as [vascular dementia](#) and [dementia with Lewy bodies](#).

Traditional diagnostic criteria for Alzheimer disease include all of the following:

- Dementia established clinically and documented by a formal mental status examination
- Deficits in ≥ 2 areas of cognition
- Gradual onset (ie, over months to years, rather than days or weeks) and progressive worsening of memory and other cognitive functions
- No disturbance of consciousness
- Onset after age 40 years, most often after age 65
- No systemic or brain disorders (eg, tumor, stroke) that could account for the progressive deficits in memory and cognition

However, deviations from these criteria do not exclude a diagnosis of Alzheimer disease, particularly because patients may have mixed dementia.

The National Institute on Aging–Alzheimer's Association diagnostic guidelines ([2](#)) also include biomarkers for the pathophysiologic process of Alzheimer disease:

- A low level of beta-amyloid in cerebrospinal fluid (CSF)
- Beta-amyloid deposits in the brain detected by positron emission tomography (PET) imaging using radioactive tracer that binds specifically to beta-amyloid plaques (eg, Pittsburgh compound B [PiB], florbetapir)

Other biomarkers indicate downstream neuronal degeneration or injury:

- Elevated levels of tau protein in CSF or tau deposits in the brain detected by PET imaging using radioactive tracer that binds specifically to tau
- Decreased cerebral metabolism in the temporoparietal cortex measured using PET with fluorine-18 (18F)-labeled deoxyglucose (fluorodeoxyglucose, or FDG)
- Local atrophy in the medial, basal, and lateral temporal lobes and the medial parietal cortex, detected by MRI

These findings increase the probability that dementia is due to Alzheimer disease. However, the guidelines (2, 4) do not advocate routine use of these biomarkers for diagnosis because standardization and availability are limited. Also, they do not recommend routine testing for the apo epsilon-4 allele.

Laboratory tests (eg, thyroid-stimulating hormone, vitamin B12 levels) and neuroimaging (MRI or CT) are done to check for other, treatable causes of dementia and disorders that can worsen symptoms. If clinical findings suggest another underlying disorder (eg, HIV, syphilis), tests for those disorders are indicated.

Differential diagnosis

Distinguishing Alzheimer disease from other dementias is difficult. The Modified Hachinski Ischemic Score is sometimes used to help distinguish [vascular dementia \(mainly multi-infarct dementia\)](#) from Alzheimer disease; it has limited clinical usefulness, but it is helpful when neuroimaging is not available (5). Key features can help distinguish Alzheimer disease from multi-infarct dementia, a type of vascular cognitive impairment and dementia (see table [Differences Between Vascular Cognitive Impairment and Dementia and Alzheimer Disease](#)). Fluctuations in cognition, parkinsonian symptoms, well-formed visual hallucinations, and relative preservation of short-term memory suggest [dementia with Lewy bodies](#) (see table [Differences Between Alzheimer Disease and Dementia with Lewy Bodies](#)).

Patients with Alzheimer disease are often better-groomed and neater than patients with other dementias.

TABLE		
Differences Between Vascular Cognitive Impairment and Dementia and Alzheimer Disease		
Clinical Evaluation	Vascular Cognitive Impairment and Dementia	Alzheimer Disease
History of present illness	Stepwise progression in cognition after multiple infarcts Subcortical ischemic vascular disease with slow progression	Slow deterioration of cognition

Clinical Evaluation	Vascular Cognitive Impairment and Dementia	Alzheimer Disease
Past medical history	Cerebrovascular risk factors or disease Coronary artery disease Peripheral vascular disease	
Family history	Vascular cognitive impairment and dementia	Alzheimer disease
Mental status	Variable, depending on location of ischemic damage Subtle decline in executive function due to subcortical ischemic vascular disease	Memory impairment typical of Alzheimer disease (eg, forgets rapidly, misses environmental and social cues)
Neurologic findings	Focal deficits, gait disturbances	Impaired mental status; otherwise normal until disease progresses
Cardiovascular findings	Findings related to cardiovascular risk factors (eg, skin changes in the legs due to peripheral artery disease, pitting edema due to heart failure, irregular heart rhythm due to atrial fibrillation, hypertension)	Usually normal
Laboratory tests needed	Measurement of hemoglobin A1C and lipids, which may be elevated ECG to check for atrial fibrillation	Routine laboratory tests to check for reversible dementia (comprehensive metabolic panel, CBC, TSH, vitamin B12 level, fluorescent treponemal antibody absorption) to rule out other possible causes of cognitive impairment

Clinical Evaluation	Vascular Cognitive Impairment and Dementia	Alzheimer Disease
MRI/CT	Focal infarcts in specific brain areas critical for cognition and behavior,, confluent white matter changes In pure vascular cognitive impairment and dementia, relative preservation of hippocampal volume	Hippocampal, temporal, parietal, and frontal atrophy
FDG-PET	Multifocal hypometabolism varying by location of vascular brain damage	Hypometabolism in parietal, temporal, and later on in frontal lobes with sparing of primary motor-sensory cortex
Biomarkers	None	Amyloid-beta or phosphorylated tau detected in CSF or by PET
CBC = complete blood cell count; CSF = cerebrospinal fluid; ECG = electrocardiography; FDG-PET = fludeoxyglucose positron emission tomography; PET = positron emission tomography; TSH = thyroid-stimulating hormone.		
Adapted from Wong EC, Chui HC : Vascular cognitive impairment and dementia. <i>Continuum</i> (Minneap Minn) 28 (3):750–780, 2022. doi: 10.1212/CON.0000000000001124		

TABLE		
Differences Between Alzheimer Disease and Dementia with Lewy Bodies		
Feature	Dementia with Lewy Bodies	Alzheimer Disease
Pathology	Lewy bodies in neurons of the cortex	Neuritic plaques, neurofibrillary tangles, and beta-amyloid deposits in the cerebral cortex and subcortical gray matter

Feature	Dementia with Lewy Bodies	Alzheimer Disease
Epidemiology	Affects twice as many men	Affects twice as many women
Inheritance	Rarely familial	Familial in 5–15% cases
Day-to-day fluctuation	Prominent	Some
Short-term memory	Less affected Deficits in alertness and attention more than in memory acquisition	Lost early in the disease
Parkinsonian symptoms	Prominent, obvious early in the disease Axial rigidity and unstable gait	Very rare, occurring late in the disease Normal gait
Autonomic dysfunction	Common	Rare
Hallucinations	Occur in about 80%, usually when disease is early Most commonly visual	Occur in about 20% of patients, usually when disease is moderately advanced
Adverse effects with antipsychotics	Common Acute worsening of extrapyramidal symptoms, which may be severe or life threatening	Common Possible worsening of symptoms of dementia

Diagnosis references

1. [NICE: National Institute for Health and Care Excellence](#): Dementia: assessment, management and support for people living with dementia and their carers. NICE guideline [NG97]. Published: 20 June 2018
2. [Jack CR Jr, Andrews JS, Beach TG, et al](#): Revised criteria for diagnosis and staging of Alzheimer's disease: Alzheimer's Association Workgroup. *Alzheimers Dement* 20(8):5143–5169, 2024. doi:10.1002/alz.13859
3. [Alzheimer's Association](#): 2018 Alzheimer's Association Dementia Care Practice Recommendations. January 18, 2028. Accessed February 3, 2025.
4. [Chételat G, Arbizu J, Barthel H, et al](#): Amyloid-PET and 18 F-FDG-PET in the diagnostic investigation of Alzheimer's disease and other dementias. *Lancet Neurol* 19:951–962, 2020. doi: 10.1016/S1474-4422(20)30314-8

5. [Hachinski VC, Iliff LD, Zilhka E, et al](#): Cerebral blood flow in dementia. *Arch Neurol* 32(9):632–637, 1975. doi: 10.1001/archneur.1975.00490510088009

Treatment of Alzheimer Disease

- Safety and support measures
- Possibly cholinesterase inhibitors and memantine
- Anti-amyloid monoclonal antibody therapy in selected patients

[Safety and supportive measures](#) for Alzheimer disease are the same as those for all dementias. For example, the environment should be bright, cheerful, and familiar, and it should be designed to reinforce orientation (eg, placement of large clocks and calendars in the room). Measures to ensure patient safety (eg, signal monitoring systems for patients who wander) should be implemented.

Providing [help for caregivers](#), who may experience substantial stress, is also important. Nurses and social workers can teach caregivers how to best meet the patient's needs. Health care professionals should watch for early symptoms of caregiver stress and burnout and, when needed, suggest support services.

Symptomatic treatment of Alzheimer disease

Cholinesterase inhibitors modestly improve cognitive function and memory in some patients. Generally, donepezil, rivastigmine, and galantamine are equally effective.

- Donepezil is a first-line medication because it has once-a-day dosing and is well-tolerated. Treatment should be continued if functional improvement is apparent after several months, but otherwise it should be stopped. The most common adverse effects are gastrointestinal (eg, nausea, diarrhea). Rarely, dizziness and cardiac arrhythmias occur. Adverse effects can be minimized by increasing the dose gradually (see table Medications for Alzheimer Disease).
- Rivastigmine is available in liquid solution and a patch.
- Galantamine modulates nicotinic receptors. It appears to stimulate release of acetylcholine and enhances its effect. Galantamine may be more beneficial for behavioral symptoms than other medications.

Memantine, an *N*-methyl-D-aspartate (NMDA) receptor antagonist, appears to improve cognition and functional capacity of patients with moderate to severe Alzheimer disease. For patients with renal insufficiency, the dose should be reduced or the medication should be avoided. Memantine can be used with a cholinesterase inhibitor.

Other medications are being studied. Efficacy of high-dose vitamin E, selegiline, nonsteroidal anti-inflammatory drugs (NSAIDs), *Ginkgo biloba* extracts, and statins is unclear. Estrogen therapy does not appear useful in prevention or treatment and may be harmful.

Disease-modifying treatment of Alzheimer disease

Anti-amyloid monoclonal antibody therapy may be appropriate for selected patients with mild disease. Anti-amyloid monoclonal antibodies demonstrate disease modification by reducing amyloid plaque burden in the brain and slowing cognitive decline, although the observed clinical benefits have been modest. The mechanism of action of anti-amyloid antibodies is to bind and remove aggregated beta-amyloid. Clinical trials of anti-amyloid monoclonal antibodies demonstrated a statistically significant reduction in the rate of clinical decline as identified by cognitive and functional outcome measures in patients with early Alzheimer disease and confirmed amyloid pathology by amyloid PET scanning or cerebrospinal fluids study.

Lecanemab is a recombinant humanized immunoglobulin gamma 1 (IgG1) anti-amyloid monoclonal antibody that binds to amyloid oligomers, protofibrils, and insoluble fibrils. In a randomized trial, patients with mild cognitive impairment and mild Alzheimer disease who received lecanemab showed a 27% reduction in cognitive decline at 18 months when compared with placebo, as measured by the Clinical Dementia Rating-Sum of Boxes (CDR-SB) score (1).

Donanemab is an IgG antibody targeting pyroglutamate on mature beta-amyloid plaques. In another randomized trial including patients with mild cognitive impairment and mild Alzheimer disease, clinical decline, as measured by the integrated Alzheimer's Disease Rating Scale (iADRS), was reduced by 35% in patients receiving donanemab compared to placebo at 18 months (2).

Anti-amyloid monoclonal antibody therapies are associated with significant adverse effects, including amyloid-related imaging abnormalities (ARIA), which consist of MRI signal changes of cerebral edema (ARIA-E) and/or microhemorrhage and superficial hemosiderosis (ARIA-H). In the aforementioned randomized trials, ARIA-E developed in 12.6% of patients receiving lecanemab and in 24% of subjects receiving donanemab (1, 2). In addition, ARIA-H occurred in 17.3% of patients receiving lecanemab and in 19.7% of those receiving donanemab. ARIA usually occurred early in treatment course and caused no symptoms for the majority of the patients. However, up to 0.8% of patients treated with lecanemab had ARIA with severe symptoms, including headache, confusion, vision disturbance, disorientation, gait disturbance, and seizures (3). There were 3 deaths among patients receiving donanemab due to severe ARIA. Serial surveillance brain MRIs have been recommended to monitor ARIA for patients receiving anti-amyloid monoclonal antibodies. Patients carrying the E4 allele of the *ApoE* gene, particularly homozygous carriers, have a significantly higher occurrence rate of ARIA (3). Testing for ApoEε4 status is recommended prior to initiation of anti-amyloid treatment to assess the risk of developing ARIA.

Treatment references

1. [van Dyck CH, Swanson CJ, Aisen P, et al](#): Lecanemab in Early Alzheimer's Disease. *N Engl J Med* 388(1):9–21, 2023. doi: 10.1056/NEJMoa2212948
2. [Sims JR, Zimmer JA, Evans CD, et al](#): Donanemab in Early Symptomatic Alzheimer Disease: The TRAILBLAZER-ALZ 2 Randomized Clinical Trial. *JAMA* 330(6):512–527, 2023. doi: 10.1001/jama.2023.13239
3. [Cummings J, Apostolova L, Rabinovici GD, et al](#): Lecanemab: Appropriate Use Recommendations. *J Prev Alzheimers Dis* 10(3):362–377, 2023. doi:10.14283/jpad.2023.30

Prognosis for Alzheimer Disease

Although progression rate varies in patients with Alzheimer disease, cognitive decline is inevitable. Survival from time of diagnosis was 9 years in a large cohort (1), although this figure varies among studies and is likely influenced by other demographic and clinical factors. Average survival from the time patients can no longer walk is about 6 months.

End-of-life issues

Because insight and judgment deteriorate in patients with dementia, appointment of a family member, guardian, or lawyer to oversee finances may be necessary. Early in dementia, before the patient is incapacitated, the patient's wishes about care should be clarified, and financial and legal arrangements (eg, durable power of attorney, [durable power of attorney for health care](#)) should be made. When these documents are signed, the patient's [capacity](#) should be evaluated, and evaluation results recorded. Decisions about artificial feeding and treatment of acute disorders are best made before the need develops.

In advanced dementia, [palliative measures](#) may be more appropriate than highly aggressive interventions or hospital care.

Prognosis reference

1. [de Melo Queiroz E, Marques Couto C, da Cruz Mecone CA, Souza Lima Macedo W, Caramelli P](#): Clinical profile and survival analysis of Alzheimer's disease patients in a Brazilian cohort. *Neurol Sci* 45(1):129-137, 2024. doi:10.1007/s10072-023-06937-z

Prevention of Alzheimer Disease

Observational evidence suggests that risk of Alzheimer disease may be decreased by the following:

- Continuing to do challenging mental activities (eg, learning new skills, doing crossword puzzles) well into old age
- Exercising regularly
- Controlling hypertension
- Lowering cholesterol levels
- Consuming a diet rich in omega-3 fatty acids and low in saturated fats

There is mixed evidence regarding the impact of alcohol on Alzheimer disease, with some studies indicating a protective effect and others suggesting an increased risk (1). Once dementia develops, abstaining from alcohol is usually recommended because alcohol can worsen dementia symptoms.

Prevention reference

1. [Piazza-Gardner AK, Gaffud TJ, Barry AE](#): The impact of alcohol on Alzheimer's disease: a systematic review. *Aging Ment Health* 17(2):133–146, 2013. doi:10.1080/13607863.2012.742488

Key Points

- Although genetic factors can be involved, most cases of Alzheimer disease are sporadic, with risk best predicted by patient age.
- Differentiating Alzheimer disease from other causes of dementia (eg, vascular dementia, dementia with Lewy bodies) can be difficult but is often best done using clinical criteria, which are 85% accurate in establishing the diagnosis.
- Treat Alzheimer disease with a combination of supportive measures and pharmacotherapy, which may include cholinesterase inhibitors, memantine, and anti-amyloid monoclonal antibody therapies.

More Information

The following English-language resource may be useful. Please note that The Manual is not responsible for the content of this resource.

[Alzheimer's Association](#)



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